CLINICAL GENOMICS & PHARMACOGENOMICS ACTION BRIEF

Patient: Melinda Ehrle (Melinda_Ehrle) · **Date:** October 13, 2026 · **Primary Audience:** Patient, Primary Care Physician (PCP) & Specialists

SECTION 1 · PRIORITY GENOMIC FINDINGS, CARRIER STATUS & CLINICAL MEANING

GENE & BIOLOGICAL SYSTEM	VARIANT & ZYGOSITY	CLASSIFICATION & CONDITION	CLINICAL IMPACT & PRACTICAL GUIDANCE
POLG **POLG** Heart & Cardiovascular	**p.Gly737Arg · c.2209G>C · rs121918054** **Heterozygous** Autosomal Dominant	**Pathogenic/Likely pathogenic** Spinocerebellar ataxia with epilepsy	Strict contraindication against valproic acid due to fatal liver toxicity risk. Carrier retains normal baseline mitochondrial function.
TAT **TAT** Multisystem	**p.Arg57Ter · c.169C>T · rs118203914** **Heterozygous** Autosomal recessive	**Pathogenic** Tyrosinemia type II	Single-copy heterozygous carrier; maintains normal enzymatic function on standard diet. No protein or tyrosine restriction required.
ATM **ATM** Multisystem	**c.1236-2del · rs1565381646** **Heterozygous** Autosomal recessive	**Pathogenic/Likely pathogenic** Hereditary cancer-predisposing syndrome	Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP.
VDR **VDR** Multisystem	**c.-1172A>G · rs4516035** **Heterozygous** Pathway Modifier	**Likely pathogenic** Chronic obstructive pulmonary disease	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
MC1R **MC1R** Multisystem	**p.Asp294His · c.880G>C · rs1805009** **Heterozygous** Pathway Modifier	**Conflicting classifications of pathogenicity|risk factor** Skin and Hair Hypopigmentation	Increased UV skin sensitivity. Annual full-body skin examination and routine broad-spectrum SPF 30+ sun protection.
AMPD1 **AMPD1** Multisystem	**p.Glu435Gly · c.1304A>G · rs764950035** **Heterozygous** Autosomal recessive	**Uncertain significance** Muscle AMP deaminase deficiency	Single-copy heterozygous carrier; normal muscle energy metabolism. Symptom-guided check only if exercise cramping or statin myalgia occurs.
CTH **CTH** Multisystem	**p.Thr67Ile · c.200C>T · rs28941785** **Heterozygous** Autosomal recessive	**Conflicting classifications of pathogenicity** CTH-related disorder	Preserves baseline homocysteine clearance. Review routine B-vitamin intake (B6, B12, folate) during annual wellness checks.

SECTION 2 · HIGH-PRIORITY MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA)

GENE & VARIANT / DIPLOTYPE	PRESCRIBING ACTION	AFFECTED MEDICATION(S)	PRESCRIBING RULE & CLINICAL GUIDANCE
POLG · p.Gly737Arg · c.2209G>C · rs121918054 (Heterozygous Carrier)	🚨 LIFE-THREATENING RISK · CONTRAINDICATED	**Valproic Acid, Sodium Valproate, Divalproex Sodium**	**CRITICAL HAZARD:** Pathogenic POLG carrier; valproate formulations are strictly contraindicated due to high risk of fatal valproate-induced hepatic necrosis / mitochondrial liver failure.
CYP2C19 · *2/*2 [splice defect (rs12769205) A>G, splice defect (rs4244285) G>A] (Poor Metabolizer)	⚠️ HIGH RISK · AVOID MEDICATION	**Clopidogrel (Plavix)**	Significantly reduced active metabolite formation and high risk of cardiovascular ischemic events/stent thrombosis. Avoid clopidogrel; prescribe alterna… Select a non-substrate alternative per CPIC.
VKORC1 · *H6/*H6 [regulatory/intronic (rs7294) C>T] (Altered Enzymatic / Pathway Activity)	⬇️ HIGH RISK · REDUCE DOSE	**Warfarin (Coumadin)**	Carries VKORC1 -1639G>A sensitivity allele (rs9923231 variant). High pharmacological sensitivity to vitamin K antagonists requiring lower initial a… Initiate at reduced starting dose per CPIC.
ABCG2 · *ref/*rs2231142 [Q141K (rs2231142) G>T] (Decreased Function)	⚠️ USE WITH CAUTION	**Antineoplastic Agents, Apixaban, Dolutegravir, Felodipine, Iguratimod, Sorafenib (+3 more)**	Genotype GG is associated with decreased response to antineoplastic agents in people with Colorectal Neoplasms as compared to genotypes GT + TT. Monitor clinical response and consider dose adjustment per CPIC.

SECTION 3 · HIGH-LEVEL PREVENTIVE CARE & ANNUAL CHECKUP DISCUSSION POINTS

GENOMIC FINDING	BIOLOGICAL PATHWAY	PREVENTIVE CARE TOPIC	CARE SETTING & TIMING	CLINICAL CONTEXT & GUIDANCE
POLG **p.Gly737Arg · c.2209G>C · rs121918054** Heterozygous (Autosomal Dominant)	**POLG · Mitochondrial DNA Maintenance & Medication Safety** Pathogenic/Likely pathogenic	**EHR Valproate Allergy/Avoidance Flag & Routine Liver Panel** Permanent Prescribing Precaution	**Primary Care Physician (PCP)** & **Prescribing Clinicians** Permanent EHR Flag & Annual Checkup	Strict contraindication against valproic acid due to fatal liver toxicity risk. Carrier retains normal baseline mitochondrial function.
TAT **p.Arg57Ter · c.169C>T · rs118203914** Heterozygous (Autosomal recessive)	**TAT · Hepatic Tyrosine Metabolism (Tyrosinemia type II)** Pathogenic	**Reassurance Only — Normal Unrestricted Diet** Optional Routine Metabolic Panel at Annual Checkup	**Primary Care Physician (PCP)** Routine Annual Checkup	Single-copy heterozygous carrier; maintains normal enzymatic function on standard diet. No protein or tyrosine restriction required.
ATM **c.1236-2del · rs1565381646** Heterozygous (Autosomal recessive)	**ATM · DNA Repair & Hereditary Cancer Screening** Pathogenic/Likely pathogenic	**Guideline-Directed Breast & Cancer Screening Discussion (per NCCN)** Preventive Screening Review	**Primary Care Physician (PCP)** & **Preventive Oncology / Genetics** Annual Preventive Screening Review	Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP.
VDR **c.-1172A>G · rs4516035** Heterozygous (Pathway Modifier)	**VDR · Vitamin D Signaling & Bone Health** Likely pathogenic	**Routine Vitamin D & Calcium Check** Annual Wellness Panel	**Primary Care Physician (PCP)** Annual PCP Checkup	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
MC1R **p.Asp294His · c.880G>C · rs1805009** Heterozygous (Pathway Modifier)	**MC1R · Skin Pigmentation & UV Sensitivity** Conflicting classifications of pathogenicity|risk factor	**Annual Skin Examination & Daily Sun Protection** Dermatology / PCP Skin Check	**Primary Care Physician (PCP)** & **Dermatology** Annual Preventive Skin Check	Increased UV skin sensitivity. Annual full-body skin examination and routine broad-spectrum SPF 30+ sun protection.

SECTION 4 · ACTIONABLE PHARMACOGENOMIC PRESCRIBING GUIDE (CPIC / DPWG)

PHARMACOGENE & DIPLOTYPE	METABOLIZER PHENOTYPE	ACTIONABLE MEDICATION SUBSTRATES	CPIC CLINICAL DOSING GUIDANCE
ABCG2 *ref/*rs2231142 Q141K (rs2231142) G>T	**Decreased Function**	**Caution/Monitor:** Antineoplastic Agents, Apixaban, Dolutegravir, Felodipine, Iguratimod	Genotype GG is associated with decreased response to antineoplastic agents in people with Colorectal Neoplasms as compared to genotypes GT + TT.
CYP2C19 *2/*2 splice defect (rs12769205) A>G, splice defect (rs4244285) G>A	**Poor Metabolizer**	**Avoid: Clopidogrel (Plavix)**	Significantly reduced active metabolite formation and high risk of cardiovascular ischemic events/stent thrombosis. Avoid clopidogrel; prescribe al…
VKORC1 *H6/*H6 regulatory/intronic (rs7294) C>T	**Altered Enzymatic / Pathway Activity**	**Reduce Dose: Warfarin (Coumadin)**	Carries VKORC1 -1639G>A sensitivity allele (rs9923231 variant). High pharmacological sensitivity to vitamin K antagonists requiring lower initial a…
CYP3A43 *2/*3 Y25fs (rs61469810) delA, P340A (rs680055) C>G	**Unknown**	Standard CPIC substrate dosing	Standard dosing guidelines apply based on diplotype.
CYP3A5 *3/*3 splice defect (rs776746) T>C	**Poor Metabolizer**	Standard CPIC substrate dosing	Standard dosing guidelines apply based on diplotype.

SECTION 5 · CLINICAL PERSPECTIVE: BALANCING PROACTIVE CARE VS. OVER-INTERVENTION

✅ Why These Findings Are Useful

- **Proactive Medication Safety:** Recording pharmacogenomic metabolizer status in the medical chart helps select effective medications and avoid preventable side effects before a prescription is started.
- **Informed Annual Checkups:** Highlighting relevant biological pathways allows the Primary Care Physician to tailor routine wellness checks and age-appropriate preventive screening.

🛑 Why Aggressive Intervention Is Not Needed

- **Single-Copy Carrier Reassurance:** No two-copy (biallelic) recessive disease genotypes are present; single-copy recessive carriers retain a functional gene copy and maintain normal baseline health.
- **Modifier Variants Are Not Diagnoses:** Common pathway modifiers indicate mild physiological tendencies and should never trigger unnecessary procedures or unprovoked medication changes.

SECTION 6 · COPY-PASTE PHYSICIAN HANDOFF / EHR CHART SUMMARY

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CLINICAL GENOMICS & PHARMACOGENOMICS SUMMARY (Melinda Ehrle | ID: Melinda_Ehrle)
• PRIORITY GENOMIC & CARRIER FINDINGS: POLG (p.Gly737Arg · c.2209G>C · rs121918054, Heterozygous), TAT (p.Arg57Ter · c.169C>T · rs118203914, Heterozygous), ATM (c.1236-2del · rs1565381646, Heterozygous), VDR (c.-1172A>G · rs4516035, Heterozygous), MC1R (p.Asp294His · c.880G>C · rs1805009, Heterozygous), AMPD1 (p.Glu435Gly · c.1304A>G · rs764950035, Heterozygous). Single-copy (heterozygous) recessive carrier states retain one functional gene copy and do not cause primary recessive disease.
• MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA):
  - AVOID / CONTRAINDICATED: Valproic Acid, Sodium Valproate, Divalproex Sodium · Clopidogrel (Plavix)
  - REDUCE STARTING DOSE: Warfarin (Coumadin)
  - RAPID METABOLISM / MONITOR: Antineoplastic Agents, Apixaban, Dolutegravir, Felodipine, Iguratimod, Sorafenib (+3 more)
• PREVENTIVE CARE PLAN: Review high-level wellness and screening topics in Section 3 during routine primary care checkups.
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